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3714 空気力を受ける共振したはりの最適設計

Optimum design of a resonating fluid-loaded beam

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Abstract

A simply supported beam of circular cross sections is assumed to be set into resonant oscillations by the action of a distributed external force. The oscillations generate sound. The aim of the present work is to determine the optimal radius distribution of the beam, which minimizes the emitted acoustic power, under constraints on the resonance frequency and beam volume. This optimization problem is solved numerically.

Keywords: Structural acoustics, resonance, fluid loading, acoustic power flow, optimum design

Introduction

The sound generation by resonant oscillations of a beam of circular cross sections in an acoustic (compressible) medium has been considered by Junger [1], Crighton [2], and Achenbach and Qu [3]. An interesting discussion in given in Ref. [2] on how the acoustic emission can be reduced when the resonant oscillations are driven by an incident pressure field. In this case the total sound pressure field consists of (*i*) the incident field, (*ii*) the scattered field, and (*iii*) the field generated by the motion of the beam. The scattered field is best reduced by reducing the product $k_0 a$, where $k_0 = \omega/c_0 =$ (frequency)/(speed of sound) is the acoustic wave number, and a is the radius of the beam. This will however increase the amplitude at resonance, and thus increase the motion-generated field.

It is thus not trivial to reduce the total sound field, and constitutes and interesting optimum design problem: What is the optimum radius distribution $a(z)$ which, for fixed volume and resonance frequency, minimizes the total sound field?

The present short note will however consider an optimum design problem related to the somewhat simpler formulation considered in Ref. [3], where the beam is driven by a 'mechanical' time-harmonic distributed load. The radius distribution which minimizes the acoustic power flow at resonance is determined with constraints on the resonance frequency and the volume of the beam.

Analysis

The present study is based on Crighton's [2] approximate analytical expression for the fluid loading, which gives the equation of motion for the fluid-loaded beam as

$$-\omega^2 m u + (EIu'')'' + \rho\pi\omega^2 a^2 \mathfrak{F}(k_0 a) u = f, \qquad (1)$$

where the frequency term $\exp(-\mathrm{i}\omega t)$ has been omitted. For Eq. (1) to be valid it is necessary that the deflection $u(z)$ varies 'slowly' with the axial position z. That is to say, (1) is only a good approximation for the lower modes of vibration. In this equation, ω is the frequency, $m(z) = \rho_s \pi a^2(z)$ is the mass distribution function with $a(z)$ being the radius of the rod and ρ_s the density of the rod material; E is Young's modulus, $I(z)$ is the area moment of inertia, ρ is the density of the surrounding fluid, and $f(z)$ is the external forcing function. A 'dash' denotes differentiation with respect to z. Finally, the complex function $\mathfrak{F}(k_0 a)$ is defined by

$$\mathfrak{F}(k_0 a) = H_1(k_0 a)/k_0 a H_1'(k_0 a), \qquad (2)$$

where $H_1 = H_1^{(1)}$ is the Hankel function of first kind and of first order. Here the 'dash' denotes differentiation with respect to the argument. It is noted that the real and imaginary parts of $\mathfrak{F}$ represent added mass and radiation damping, respectively.

Considering a simply supported beam, the boundary conditions are $u = 0$, $u'' = 0$, at $z = 0, L$.

Scaling of the variables as

$$\bar{u} = u/L, \quad \bar{z} = z/L, \quad \bar{a} = a/a_0, \quad \kappa = k_0 a_0, \qquad (3)$$
$$\epsilon = \frac{\rho}{\rho_s}, \quad \bar{\omega} = \omega L^2 \left(\frac{4\rho_s}{E a_0^2}\right)^{\frac{1}{2}}, \quad \bar{f} = \frac{4}{\pi}\frac{L^3}{a_0^4 E} f,$$

where a_0 is a reference radius, gives the non-dimensional problem

$$-\omega^2 a^2 u + (a^4 u'')'' + \epsilon \mathfrak{F}(\kappa a)\omega^2 a^2 u = f, \qquad (4)$$
$$u(0) = u''(0) = u(1) = u''(1) = 0,$$

where the 'overbars' have been dropped again for sake of simple notation.

The system (4) is discretized by using the Galerkin finite element method, which transforms (1) into the non-homogeneous linear equation system

$$\mathbf{K}(\omega)\mathbf{d} = \left[-\omega^2 \mathbf{M} + \mathbf{S} + \omega^2 \mathbf{D}(\omega)\right] \mathbf{d} = \mathbf{f}(\omega). \qquad (5)$$

Within each of the J_e elements, the the element beam radius a_j is taken to be constant. The dependence of the element matrices $\mathbf{M}_j$, $\mathbf{S}_j$, $\mathbf{D}_j$ upon the design parameters a_j can be written as $\mathbf{M}_j = a_j^2 \bar{\mathbf{M}}_j$, $\mathbf{S}_j = a_j^4 \bar{\mathbf{S}}_j$, and $\mathbf{D}_j = a_j^2 \mathfrak{F}(\kappa a_j)\bar{\mathbf{D}}_j$, respectively, where a bar denotes a constant matrix.

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The acoustic power flow is given by

$$\Pi = \tfrac{1}{2}\omega \left|\mathrm{Im}\left(f\, u^*\right)\right|, \tag{6}$$

where an asterisk stands for complex conjugation. In terms of the discretized form (5), Eq. (6) is written as $\Pi = \frac{1}{2}\omega \left|\mathrm{Im}\left(\mathbf{f}^T\mathbf{d}^*\right)\right|$, where a superscript T stands for transposition.

The n'th resonance frequency ω_n is determined as the n'th smallest eigenvalue of the real eigenvalue problem $\left[-\omega^2\mathbf{M} + \mathbf{S} + \omega^2\mathrm{Re}\{\mathbf{D}(\omega)\}\right] = \mathbf{0}$.

Optimization problem

The optimization problem is to determine the set of element beam radii $\{a_1, a_2, \ldots, a_{J_e}\}$ which, at the n'th resonance frequency ω_n,

$$\text{minimize } \Pi \tag{7}$$

subject to

$$\begin{aligned} V_{\min} &\le V \le V_{\max}, \\ \omega_{0\,\min} &\le \omega_n \le \omega_{0\,\max}, \\ a_{\min} &\le a_j \le a_{\max}, \quad j = 1, 2, \cdots, J_e \end{aligned} \tag{8}$$

where V is the volume of the beam. It would be desirable to solve (7)-(8) with both volume and resonance frequency kept constant. But the resonance frequency is rather sensitive to design changes, so some 'slack' on these variables is necessary.

In order to solve (7)-(8), the derivatives $\partial\Pi/\partial a_j$ need to be determined. Using the so-called adjoint method it can be shown that

$$\frac{\partial \Pi}{\partial a_j} = \tfrac{1}{2}\omega_n \,\mathrm{Im}\left(\frac{\partial \mathbf{f}^T}{\partial a_j}\mathbf{u}^* + \mathbf{\Lambda}^T\frac{\partial \mathbf{f}^*}{\partial a_j} - \mathbf{\Lambda}^T\frac{\partial \mathbf{K}^*}{\partial a_j}u^*\right)\hat{s},$$

where $\hat{s} = \mathrm{sgn}\left\{\mathrm{Im}\left(\mathbf{f}^T\mathbf{u}^*\right)\right\}$. The vector $\mathbf{\Lambda}$ is determined by solving the to (5) 'adjoint problem' $\mathbf{K}^*\mathbf{\Lambda} = \mathbf{f}$.

To keep the resonance frequency ω_n within the specified bounds, the derivative of this function with respect to a_j is also needed, and is given by

$$\partial\omega_n/\partial a_j = -N_j/D_j, \tag{9}$$

where $N_j = \mathbf{d}_j^T\left\{-\omega_n^2\frac{\partial \mathbf{M}}{\partial a_j} + \frac{\partial \mathbf{S}}{\partial a_j} + \omega_n^2\frac{\partial \mathbf{D}(\omega_n)}{\partial a_j}\right\}\mathbf{d}_j$ and $D_j = 2\omega_n\mathbf{d}_j^T\left\{-\mathbf{M} + \mathbf{D}(\omega_n)\right\}\mathbf{d}_j + \omega_n^2\mathbf{d}_j^T\frac{\partial \mathbf{D}(\omega_n)}{\partial \omega_n}\mathbf{d}_j$.

The optimization problem (7-8) is solved numerically using the 'Method of Moving Asymptotes' (MMA) [4].

Numerical example

The numerical examples shown in Fig. 1, obtained using 60 finite elements, are for a PVC rod in water. The forcing function $f(z) = \gamma \sin n\pi z$, with γ being a constant, and with n corresponding to the n'th resonance frequency, ω_n.

In terms of the non-dimensional formulation, the initial design has volume $V = 1.0$, $a_j \equiv 1$, and $\gamma = 0.05$. The constraint-inequalities in (8) are set as follows: $0.975 \le V \le 1.025$, $\omega_{0n} - 0.025 \le \omega_n \le \omega_{0n} + 0.025$, and $0.1 \le a_j \le 2.0$. Here ω_{0n} is the n'th resonance frequency of the initial (uniform) design.

Optimum beams and design histories for the acoustic power flow Π are shown in Fig. 1 for the cases of first ($n = 1$) and third ($n = 3$) resonance frequency. The resonance frequency ω_n for each 'intermediate' design is traced at each design step, such that Π always is evaluated at the *actual* resonance peak.

For the case $n = 1$, the power flow from the optimum beam is 87.2 % of the original level; for $n = 3$ it is 81.0 %.

As a concluding remark its is noted that the 'opposite' problem of (7), that is, to maximize Π under the same constraints, also is of technical interest, for example in the design of more efficient string instruments.

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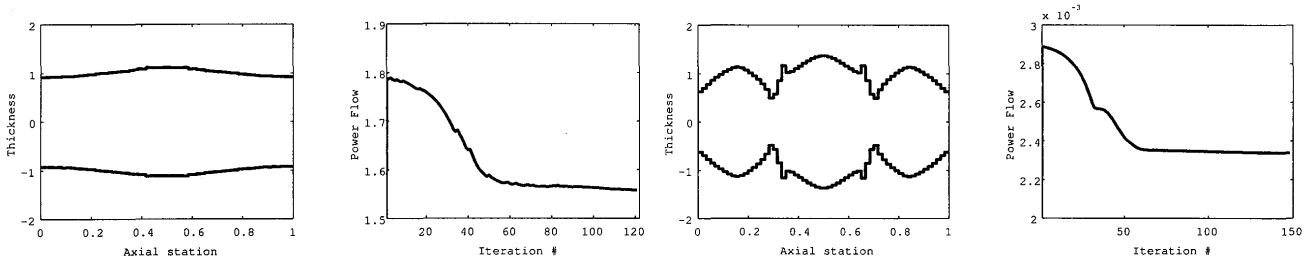
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(a) (b) (c) (d)

Figure 1: (a) Optimum design for minimum acoustic power flow at first resonance frequency. (b) Design history of the acoustic power flow by first resonance frequency. (c) Optimum design at third resonance frequency. (d) Design history by third resonance frequency.

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